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Acute Pain Management Drug Library Policy			
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1.0	16.3.17	Final	Minor amendments. Approved through Drugs and therapeutics committee
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1. Purpose

1.1. The use of drug libraries has been highlighted as an important development within the administration of intravenous drugs as they have the potential to reduce drug error caused by miscalculation of drip rates, infusion rates and healthcare professional inputting errors.

1.2. This policy enables all clinical staff within Northern Devon Healthcare NHS Trust (NDHT) to safely and effectively use intravenous infusion device inclusive of an Acute Pain Management drug library. This policy is to be used in accordance with the Trust's Injectable Medicines Policy 2016 and the Nursing & Midwifery Council Standards for Medicines Management (2007).

1.3. The policy applies to registered nurses, ODPs, anaesthetists and midwives working with Patient Controlled Analgesia(PCA) and epidural analgesia at Northern Devon Healthcare Trust

1.4. Implementation of this policy will ensure that:

- Patient safety is maintained during the administration of intravenous pain medication
- Provide guidance to healthcare professionals involved in the revalidation and development of drug libraries
- Adequate training for all staff using the drug library

2. Definitions

EBME

2.1. Electro-Biomedical Engineering

3. Responsibilities

3.1. Each of the following individuals and groups has a responsibility in the delivery of this policy.

Role of Anaesthetic Clinical Lead for Acute Pain & Anaesthetic Obstetric Lead (Maternity Epidural Programmes)

3.2. The Anaesthetic lead for Acute Pain is responsible for:

- Acting as an expert within the field of pain medication administration and provides ultimate sign-off for any drug library to be included within the infusion devices
- Approve new drug libraries for upload
- Review and approve existing drug libraries

Role of Clinical Nurse Specialist for Acute Pain

3.3. The clinical nurse specialist for acute pain is responsible for:

- Acting an expert within the field of pain medication administration and provides ultimate sign-off for any drug library to be included within the infusion devices
- Approve new drug libraries for upload
- Review and approve existing drug libraries
- Facilitating the delivery of education to users of infusion devices and when necessary, organise training on specific devices used in their department
- Assess risk and act on findings as appropriate
- Monitoring incident and investigate where appropriate
- Implementing any change in practice required to maintain patient safety

Role of Surgical Lead Pharmacist

3.4. The Lead Pharmacist is responsible for:

- Approve new drug libraries for upload
- Review and approve existing drug libraries

Role of Electro-BioMedical Engineering Manager

3.5. The Electro-BioMedical Engineering Manager is responsible for:

- Ensuring the commissioning, servicing (unless under warrantee or contract) and general upkeep of the infusion devices held within NDHT
- Upload any new or existing drug library to the infusion devices following ratification

Role of Registered Nurse and Midwives (Users of PCA/epidurals)

3.6. The registered nurses and midwives are responsible for:

- Ensuring that policy and procedure is followed when using the infusion device drug library
- Maintaining current competence in the administration of pain medication, infusion devices and drug libraries
- Maintaining patient safety at all times

4. Training

4.1. All registered professionals before using the drug library must be first deemed competent. Competence must be gained through practical assessment facilitated by the designated key trainer or training representative from CME or cardinal health.

4.2. Training will be delivered through in depth demonstration followed by assessment in practice and completion of relevant e-learning modules.

4.3. Records of assessment and competence must be signed by both the key trainer and trainee. Records will be then kept by the relevant manager of the competent individual. Competencies documents can be accessed via BOB.

5. Incidents

5.1. Any incidents during the use of the drug library must be reported to the Clinical Nurse Specialist for Acute Pain and ensuring the incident is formally logged on the Trust's electronic incident reporting system

5.2. Any pump involved with any incident must be removed from service, decontaminated and reported to the EBME department for analysis.

5.3. Incidents must be audited by the Acute Pain Management Team, identifying the level of risk and act appropriately to preserve patient safety.

6. Drug Library creation, review and upload

6.1. Each drug within the library must be written, approved and signed off by the Anaesthetic Lead for Acute Pain, Clinical Nurse Specialist for Acute Pain and the Lead surgical Pharmacist using the "drug library upload form" template in appendix 1 (IVAC Alaris for IV PCA) appendix 2 (CME Bodyguard for Epidural)

6.2. Once approved by all relevant parties, the library will be forward to EBME for upload.

6.3. Upload of the Drug Library will be completed by the EBME department and may be achieved directly through physical connection to the machine or wirelessly via a dedicated network. At point of upload, the engineer conducting the action will use the "Individual Device Drug Library Update Record" (appendix 3), to record completion of the transfer of information. Copies of the completed form will be held by EBME, and Clinical Nurse Specialist for Acute Pain.

6.4. The entire drug library will be reviewed and ratified by the Clinical Nurse Specialist for Acute Pain, Anaesthetic Lead for Acute Pain and lead surgical pharmacist on an annual basis, authorising additional drugs for upload, or sanctioning the archiving of unused drugs.

7. Use of Drug Library

7.1. Authority to use the drugs library/select alter programme parameters with agreed pre-set limits for PCA and Epidural is limited to the Acute Pain Management team, Anaesthetic team, and Theatre recovery staff.

7.2. Two registrants who are deemed competent must check the drug(s) to be administered. This must take place at the bed side with involvement of the patient where possible. The following checks should be made prior to the set up and connection of the infusion and at each syringe/infusion bag change.

- Full name
- Date of birth
- Hospital number
- Individual drug identification and doses
- Diluent and dilution volumes
- Expiry date, both treatment and diluent
- Administration route and duration
- Regimen (concentration, infusion rate, lock out period, dose duration, PCA bolus dose)

7.3. In addition Nurses, Midwives and ODPs caring for patients with PCA or Epidurals are expected to complete the following infusion device checks 4 hourly.

- Regimen (drug, concentration, infusion rate, lock out period, dose duration, PCA bolus dose). To be checked against the infusion device and prescription chart.

7.4. The drug library is accessed through both IVAC Alaris P5000 (PCA) and CME Bodyguard (Epidural) pumps via an on screen prompt. Authority to use the drugs library/select alter programme parameters for PCA and Epidural is limited to the Acute Pain Management team, Anaesthetic team, and Theatre recovery staff who have been assessed as competent.

7.5. The drug library must be utilised if the drug is listed for use. The authorised users can alter drug programme parameters in accordance with patients changing needs and in discussion with the anaesthetic and/or acute pain team.

7.6. If an error occurs relating to the set-up of the pump, the error will be classed as a medication error. An electronic incident report must be submitted and the Trust's "Drug Error Matrix" must be implemented by the investigating officer with the user.

8. Monitoring Compliance with and the Effectiveness of the Policy

Standards/ Key Performance Indicators

8.1. Key performance indicators comprise:

- Area specific training and competence for Medical Devices
- Monitoring incidents, investigations and feedback

9. Equality Impact Assessment

9.1. The author must include the Equality Impact Assessment Table and identify whether the policy has a positive or negative impact on any of the groups listed. The Author must make comment on how the policy makes this impact.

Table 1: Equality impact Assessment

Group	Positive Impact	Negative Impact	No Impact	Comment
Age			x	
Disability			x	
Gender			x	
Gender Reassignment			x	
Human Rights (rights to privacy, dignity, liberty and non-degrading treatment)			x	
Marriage and civil partnership			x	
Pregnancy			x	
Maternity and Breastfeeding			x	
Race (ethnic origin)			x	
Religion (or belief)			x	
Sexual Orientation			x	

10. Associated Documentation

- Injectable Medicines Policy 2016
- Nursing & Midwifery Council Standards for Medicines Management (2007)
- Pain Management Guidelines 201

Appendix 1 - Drug Library Update Form IVAC Alaris P5000

Drug Library reference No:

Infusion device IVAC Alaris P5000

Drug 1	**Morphine**	Safety limits	Change required (Y / N)
	Minimum: Drug conc.		
	Maximum: Drug conc.		
	Minimum: Lockout period		
	Maximum: Lockout period		
	Minimum: Dose Duration		
	Maximum: Dose Duration		
	Minimum PCA Dose		
	Maximum: PCA Dose		
	Maximum: continuous		
	Maximum: loading dose		
	Maximum: Clinician bolus		
	Default setting: Drug concentration		
	Default setting: PCA Dose		
	Default setting: Lockout Period		
	Default setting: Dose Duration		
	Default setting: Occlusion Level		
	Default setting: Continuous Rate		
	Default setting: Loading Dose		

Drug 2	**Fentanyl**	Safety limits	Change required (Y / N)
	Minimum: Drug conc.		
	Maximum: Drug conc.		
	Minimum: Lockout period		
	Maximum: Lockout period		
	Minimum: Dose Duration		
	Maximum: Dose Duration		
	Minimum PCA Dose		
	Maximum: PCA Dose		
	Maximum: continuous		
	Maximum: loading dose		
	Maximum: Clinician bolus		
	Default setting: Drug concentration		
	Default setting: PCA Dose		
	Default setting: Lockout Period		
	Default setting: Dose Duration		
	Default setting: Occlusion Level		
	Default setting: Continuous Rate		
	Default setting: Loading Dose		

Drug Library reference No:

Infusion device IVAC Alaris P5000

Drug 3	**Oxycodone**	Safety limits	Change required (Y / N)
	Minimum: Drug conc.		
	Maximum: Drug conc.		
	Minimum: Lockout period		
	Maximum: Lockout period		
	Minimum: Dose Duration		
	Maximum: Dose Duration		
	Minimum PCA Dose		
	Maximum: PCA Dose		
	Maximum: continuous		
	Maximum: loading dose		
	Maximum: Clinician bolus		
	Default setting: Drug concentration		
	Default setting: PCA Dose		
	Default setting: Lockout Period		
	Default Setting: Dose Duration		
	Default setting: Occlusion Level		
	Default setting: Continuous Rate		
	Default setting: Loading Dose		
Drug 4	**PCA Morphine Paeds**	Safety limits	Change required (Y / N)
	Minimum: Drug conc.		
	Maximum: Drug conc.		
	Minimum: Lockout period		
	Maximum: Lockout period		
	Minimum: Dose Duration		
	Maximum: Dose Duration		
	Minimum PCA Dose		
	Maximum: PCA Dose		
	Maximum: continuous		
	Maximum: loading dose		
	Maximum: Clinician bolus		
	Default setting: Drug concentration		
	Default setting: PCA Dose		
	Default setting: Lockout Period		
	Default setting: Occlusion Level		
	Default setting: Continuous Rate		
	Default setting: Loading Dose		

Drug Library reference No:

Infusion device IVAC Alaris P5000

Drug 5	**PCA Oxycodone paeds**	Safety limits	Change required (Y / N)
	Minimum: Drug conc.		
	Maximum: Drug conc.		
	Minimum: Lockout period		
	Maximum: Lockout period		
	Minimum: Dose Duration		
	Maximum: Dose Duration		
	Minimum PCA Dose		
	Maximum: PCA Dose		
	Maximum: continuous		
	Maximum: loading dose		
	Maximum: Clinician bolus		
	Default setting: Drug concentration		
	Default setting: PCA Dose		
	Default setting: Lockout Period		
	Default setting: Occlusion Level		
	Default setting: Continuous Rate		
	Default setting: Loading Dose		

Drug 5	**NCA Morphine**	Safety limits	Change required (Y / N)
	Minimum: Drug conc.		
	Maximum: Drug conc.		
	Minimum: Lockout period		
	Maximum: Lockout period		
	Minimum: Dose Duration		
	Maximum: Dose Duration		
	Minimum PCA Dose		
	Maximum: PCA Dose		
	Maximum: continuous		
	Maximum: loading dose		
	Maximum: Clinician bolus		
	Default setting: Drug concentration		
	Default setting: PCA Dose		
	Default setting: Lockout Period		
	Default setting: Occlusion Level		
	Default setting: Continuous Rate		
	Default setting: Loading Dose		

Drug Library reference No:

Infusion device IVAC Alaris P500

Drug 5	**NCA Oxycodone**	Safety limits	Change required (Y / N)
	Minimum: Drug conc.		
	Maximum: Drug conc.		
	Minimum: Lockout period		

	Maximum: Lockout period		
	Minimum: Dose Duration		
	Maximum: Dose Duration		
	Minimum PCA Dose		
	Maximum: PCA Dose		
	Maximum: continuous		
	Maximum: loading dose		
	Maximum: Clinician bolus		
	Default setting: Drug concentration		
	Default setting: PCA Dose		
	Default setting: Lockout Period		
	Default setting: Occlusion Level		
	Default setting: Continuous Rate		
	Default setting: Loading Dose		

Agreement for upload (sign-off)		
Anaesthetic Lead Acute Pain: Anaesthetic Obstetric Lead:	Signature:	
	Print Name:	
	Date:	
Clinical Nurse Specialist Acute Pain:	Signature:	
	Print Name:	
	Date:	
Lead Surgical Pharmacist:	Signature:	
	Print Name:	
	Date:	
Complete System Upload Details		
EBME	Signature:	
	Print Name:	
	Date:	

Appendix 2 - Drug Library Update Form CME Bodyguard 545

Drug Library reference No. **……..……**

Infusion device **CME Medical Bodyguard 545**

Protocol 1	**A:EPCA Fentanyl & Bupivacaine**	Safety limits	Change required (Y / N)
	Mediguard	Off	
	Min Patient Weight	0 Kg	
	Max Patient Weight	0 Kg	
	Bag Volume	490 ml	
	Concentration	mL Only	
	Initial Rate	6 ml/hr	
	Maximum Rate	12 ml/hr	
	Bolus Type	Patient Bolus	
	Initial Bolus	4 ml	
	Max Bolus	6 ml	
	Initial Lockout	20 min	
	Limit	0 ml/24hr	

Protocol 2	**B:EPCA Bupivacaine**	Safety limits	Change required (Y / N)
	Mediguard	Off	
	Min Patient Weight	0 Kg	
	Max Patient Weight	0 Kg	
	Bag Volume	490 ml	
	Concentration	mL Only	
	Initial Rate	6 ml/hr	
	Maximum Rate	12 ml/hr	
	Bolus Type	Patient Bolus	
	Initial Bolus	4 ml	
	Max Bolus	6 ml	
	Initial Lockout	20 min	
	Limit	0 ml/24hr	

Drug Library reference No. **……..……**

Infusion device **CME Medical Bodyguard 545**

Protocol 3	**C:Continuous Nerve**	Safety limits	Change required (Y / N)
	Mediguard	Off	
	Min Patient Weight	0 Kg	
	Max Patient Weight	0 Kg	
	Bag Volume	490 ml	

	Concentration	mL Only	
	Initial Rate	5 ml/hr	
	Maximum Rate	15 ml/hr	
	Bolus Type	Patient Bolus	
	Initial Bolus	0 ml	
	Max Bolus	0 ml	
	Initial Lockout	0 min	
	Limit	0 ml/24hr	

Protocol 4	**D:Obstetric PCEA 1**	Safety limits	Change required (Y / N)
	Mediguard	Off	
	Min Patient Weight	0 Kg	
	Max Patient Weight	0 Kg	
	Bag Volume	90 ml	
	Concentration	mL Only	
	Initial Rate	8 ml/hr	
	Maximum Rate	15 ml/hr	
	Bolus Type	Patient Bolus	
	Initial Bolus	5 ml	
	Max Bolus	10 ml	
	Initial Lockout	15 min	
	Limit	0 ml/24hr	

Protocol 5	**E:Obstetric PCEA 2**	Safety limits	Change required (Y / N)
	Mediguard	Off	
	Min Patient Weight	0 Kg	
	Max Patient Weight	0 Kg	
	Bag Volume	90 ml	
	Concentration	mL Only	
	Initial Rate	5 ml/hr	
	Maximum Rate	10 ml/hr	
	Bolus Type	Patient Bolus	
	Initial Bolus	8 ml	
	Max Bolus	10 ml	
	Initial Lockout	15 min	
	Limit	0 ml/24hr	

Drug Library reference No. **……..……**

Infusion device **CME Medical Bodyguard 545**

Protocol 6	**F:Obstetric PCEA 3**	Safety limits	Change required (Y / N)
	Mediguard	Off	
	Min Patient Weight	0 Kg	
	Max Patient Weight	0 Kg	

	Bag Volume	90 ml	
	Concentration	mL Only	
	Initial Rate	0 ml/hr	
	Maximum Rate	10 ml/hr	
	Bolus Type	Patient Bolus	
	Initial Bolus	10 ml	
	Max Bolus	10 ml	
	Initial Lockout	15 min	
	Limit	0 ml/24hr	

Agreement for upload (sign-off)		
Anaesthetic Lead Acute Pain: Anaesthetic Obstetric Lead:	Signature:	
	Print Name:	
	Date:	
Clinical Nurse Specialist Acute Pain:	Signature:	
	Print Name:	
	Date:	
Lead Surgical Pharmacist:	Signature:	
	Print Name:	
	Date:	
Complete System Upload Details		
EBME	Signature:	
	Print Name:	
	Date:	

Appendix 3 – Individual Device Drug Library Update Record – EBME ONLY

Model	Equip. Type	Equip. No.	Upload Date	Upload Time	Date	Print Name	Signature

IVAC Alaris P5000	Infusion Dev., Syringe	5944		:			
IVAC Alaris P5000	Infusion Dev., Syringe	5945		:			
IVAC Alaris P5000	Infusion Dev., Syringe	5946		:			
IVAC Alaris P5000	Infusion Dev., Syringe	5947		:			
IVAC Alaris P5000	Infusion Dev., Syringe	5948		:			
IVAC Alaris P5000	Infusion Dev., Syringe	5949		:			
IVAC Alaris P5000	Infusion Dev., Syringe	7171		:			
IVAC Alaris P5000	Infusion Dev., Syringe	7173		:			
IVAC Alaris P5000	Infusion Dev., Syringe	7174		:			
IVAC Alaris P5000	Infusion Dev., Syringe	7175		:			
IVAC Alaris P5000	Infusion Dev., Syringe	7176		:			
IVAC Alaris P5000	Infusion Dev., Syringe	7177		:			
IVAC Alaris P5000	Infusion Dev., Syringe	7178		:			
IVAC Alaris P5000	Infusion Dev., Syringe	7179		:			
IVAC Alaris P5000	Infusion Dev., Syringe	8311		:			
IVAC Alaris P5000	Infusion Dev., Syringe	8312		:			
IVAC Alaris P5000	Infusion Dev., Syringe	8313		:			
IVAC Alaris P5000	Infusion Dev., Syringe	8314		:			
IVAC Alaris P5000	Infusion Dev., Syringe	8315		:			
IVAC Alaris P5000	Infusion Dev., Syringe	8316		:			

Model	Equip. Type	Equip. No.	Upload Date	Uploa d Time	Date	Print Name	Signature

IVAC Alaris P5000	Infusion Dev., Syringe	8317		:			
IVAC Alaris P5000	Infusion Dev., Syringe	8318		:			
IVAC Alaris P5000	Infusion Dev., Syringe	8319		:			
IVAC Alaris P5000	Infusion Dev., Syringe	8320		:			
IVAC Alaris P5000	Infusion Dev., Syringe	8321		:			
IVAC Alaris P5000	Infusion Dev., Syringe	8322		:			
IVAC Alaris P5000	Infusion Dev., Syringe	8323		:			
IVAC Alaris P5000	Infusion Dev., Syringe	8324		:			
IVAC Alaris P5000 **PAED**	Infusion Dev., Syringe	8325		:			
IVAC Alaris P5000 **PAED**	Infusion Dev., Syringe	8326		:			
CME Bodyguard	Infusion Device epidural	10252		:			
CME Bodyguard	Infusion Device epidural	10253		:			
CME Bodyguard	Infusion Device epidural	10254		:			
CME Bodyguard	Infusion Device epidural	10255		:			
CME Bodyguard	Infusion Device epidural	10256		:			
CME Bodyguard	Infusion Device epidural	10257		:			
CME Bodyguard	Infusion Device epidural	10258		:			
CME Bodyguard	Infusion Device epidural	10259		:			
CME Bodyguard	Infusion Device epidural	10260		:			

CME Bodyguard	Infusion Device epidural	10261		:			